Assignment 3: Database Design Analysis

COLLEGE MANAGEMENT SYSTEM

DBMS Assignment 3

Database: CollegeManagementSystem

SQL Platform: MySQL 8.0.16 or later

Project scope: departments, faculty, students, courses, and enrollments.

Note: Query outputs described in this report are expected results from the included fictional sample data. Run the SQL script in MySQL Workbench and capture actual screenshots if your instructor requires them.

Purpose

Analyze redundancy, functional dependencies, insertion/deletion/update anomalies, and improvements in the College Management System.

Schema

The normalized design contains Departments, Faculty, Students, Courses, and Enrollments. Enrollments resolves the many-to-many relationship between students and courses.

Functional dependencies

Departments: department_id → department_name, office_location; department_name → department_id, office_location.
Faculty: faculty_id → faculty_name, email, department_id, salary, employment_status; email → faculty_id and other faculty attributes.
Students: student_id → register_no, student_name, email, date_of_birth, admission_year, department_id; register_no and email are alternate candidate keys.
Courses: course_id → course_code, course_name, credits, department_id, faculty_id; course_code → other course attributes.
Enrollments: enrollment_id → student_id, course_id, semester, academic_year, grade, enrollment_date; (student_id, course_id, semester, academic_year) → grade, enrollment_date.

Anomalies in a poor design

If all student, department, course, faculty, and grade details are stored in one large table, department address and faculty email repeat for every related record. Insertion anomaly: a department cannot be recorded until a student exists. Update anomaly: a faculty email change must be repeated across many rows. Deletion anomaly: deleting the last student in a department might accidentally remove the only copy of department details.

Improvements and normalization

Separate entities into tables; define primary and foreign keys; use UNIQUE constraints for alternate keys; use Enrollments as a bridge table; avoid storing derived totals repeatedly. The design targets Third Normal Form (3NF): non-key attributes depend on the key, the whole key, and nothing but the key.

Relationships

Departments 1-to-many Students, Faculty, and Courses. Faculty 1-to-many Courses. Students many-to-many Courses through Enrollments.

End of Assignment Report